🚀 Nexus Institutional Trading Terminal (Frontend)

A flagship, ultra-premium cryptocurrency trading dashboard designed with **Glassmorphism 3.0** and lightning-fast **60FPS React DOM Rendering** for high-volume traders.

🌌 Overview

The **Nexus Mini-Trading-System** is a highly reactive, responsive, and performance-tuned frontend built. Our goal was to replicate the intensity and precision of real Wall Street trading terminals (like Bloomberg and Kraken Pro), while injecting a sleek, modern, cyberpunk-inspired UI.

🎨 The Glassmorphic Grid Architecture

Unlike standard web apps, this frontend utilizes a completely custom **CSS-driven cursor-tracking system**. We bypassed native React `useState` tracking for background glows to eliminate render cycle lag. Instead, the UI reads directly from a React `useRef` and manipulates custom CSS variables (`--mouse-x`, `--mouse-y`) to power massive, full-screen transparent glowing grids—operating at a flawless 60 FPS under heavy chart loads without triggering a single React re-render.

⚡ Core Features

- **Dynamic Theme Profiles**: Every module has a distinct institutional color identity.
  - 🟦 **Dashboard**: Deep Sapphire Blue for broad portfolio overviews.
  - 🟩 **Market Heatmap**: Emerald Green for high-volume liquidity analysis.
  - 🟨 **Execution Trade Deck**: Warning Amber to signify active financial commitment.
  - 🟪 **Notification Center**: Neon Violet for critical system and security alerts.
- **Pro-Level 3-Column Trade Deck**: A complete overhaul of the trading interface replicating elite exchanges.
  - Left: Interactive Market Pairs.
  - Center: Live Rechart area graph mapping to asset volatility.
  - Right: Execution payload form & Free Margin limits.
- **Global Command Palette**: Hit `Ctrl + K` (or via the UI) to summon a global routing search bar that mimics Mac's Spotlight and immediately teleports you to deep application routes.
- **State Persistence**: Redux slices gracefully manage Session States, Portfolio Holdings, and Trade Histories, backing directly onto a Python FastAPI backend.

🛠️ Tech Stack

- **React 18** - UI Component Engine.

- **Tailwind CSS V3** - Utility-first styling with deeply configured blur drop-shadows and mesh gradients.
- **Redux Toolkit** - Bulletproof state management for Portfolio sync and Auth.
- **React Router DOM V6** - Fast, lazy-loaded Single Page Application execution.
- **Recharts** - Lightning-fast SVG charting data visualization.
- **Lucide React** - Crisp, featherweight SVG icon taxonomy.

🗺 Application Topology & Routes

Route Protocol	Terminal Module	Description
`/`	**Dashboard**	The command center. Displays total equity, active positions, and high-level spark metrics.
`/market`	**Market Intel**	Live data tables mapping 24h Trends, Volume caps, and direct 'Execute' entry points.
`/trade/:symbol`	**Execution Deck**	Real-time AreaCharts, slippage protection, and Buy/Sell execution payloads.
`/portfolio`	**Asset Matrix**	Full breakdown of holdings, unrealized P&L, pie-allocations, and withdraw portals.
`/orders`	**Transaction Ledger**	The immutable record of all historical filled and rejected orders.
`/settings`	**Security Hub**	2FA, Biometrics, API Rate limits, Session revocation, and KYC profiles.
`/support`	**Knowledge Base**	Tier-1 support pipeline, Live System Status, FAQ accordions, and Ticket Submission.
`/notifications`	**Alert Center**	Real-time push notifications for Stop-Loss triggers, Deposits, and Security warnings.

🚀 Getting Started

1. Requirements

Ensure you have **Node.js (v16+)** installed on your machine.

2. Installation

Clone the repository, drop into the directory, and invoke the install protocol:

```
npm install
```

3. Launching Development Server

Fire up Vite's HMR server:

```
npm run dev
```

The application will spin up at `http://localhost:5173`.

> **Note on Backend**: The frontend fetches live endpoints from the deployed backend instance `https://mini-trading-system-backend.onrender.com`. Ensure you have active internet connectivity to fetch live auth schemas and execution chains.

👨‍💻 Architects

Proudly engineered, refined, and deployed by:

- **Aayush**
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Built for the future of decentralized and institutional finance.